

Malcolm Zartman

☎ (XXX) XXX-XXXX | ✉ mjzartman@gmail.com | 🔗 linkedin.com/in/malcolm-zartman | 📍 San Diego, CA

Experience

Junior Data Scientist – HealthSpanIQ LLC – *San Diego, CA* *May 2026 – Present*
Health-tech startup leveraging multi-dimensional data to provide personalized insights for longevity and well-being

- Designed an automated feature selection pipeline using **RFECV**, **LightGBM**, and **SHAP** values, reducing feature space by **over 95%** (20,000+ variables to key predictors) across **500,000+** clinical records to accelerate domain research
- Engineered an automated data engineering pipeline in **Python (Pandas, NumPy, SciPy)**, cutting dataset integration time from 2 weeks to minutes via **modular ETL**, schema normalization, provenance tracking, and feature ablation testing
- Earned executive commendation for onboarding cross-functional talent and clinical experts, establishing **HIPAA / NIH data governance** standards, and replacing threshold cutoffs with continuous actuarial hazard models

Data Scientist Internship – HealthSpanIQ LLC – *San Diego, CA* *Dec 2025 – May 2026*

- Architected a proprietary, **pathology-agnostic** health-scoring engine trained on **200,000+** clinical records, delivering a full-stack MVP in under 30 days for investor demos by containerizing a REST API (**Flask, Docker**), configuring **CI/CD** pipelines (GitHub Actions) for automated testing, and deploying on **Azure / Render**
- Constructed **Databricks SQL** queries to aggregate clinical data, applying unsupervised learning (**K-Means, PCA, Silhouette Analysis**) for patient risk profiling and sequence mining to extract drug interactions backed by Ivy League research

Software Developer Internship – VillageCore – *San Diego, CA* *May 2024 – July 2024*
An AgeTech non-profit dedicated to empowering the senior community and supporting caregivers through digital tools

- Developed a multi-source data ingestion pipeline in **Python (BeautifulSoup)**, automating catalog scraping across 10+ partner networks into structured **Excel** reports to eliminate manual research and cut discovery time from **days to minutes**
- Implemented a responsive WordPress UI components using **HTML, CSS, and JavaScript** in **Agile** sprints, optimizing senior accessibility and layout navigation to support post-launch traffic and drive web app adoption

Education

Bachelor of Science in Computer Science | 3.95 / 4.00 GPA *Graduated May 2026*

California State University San Marcos – *San Marcos, CA*

- Honors & Coursework:** Graduated Summa Cum Laude, Deans List (2022–2026), International Collegiate Programming Contest (ICPC) training team | Deep Learning, Data Mining, Cloud Computing

Projects

Synthetic Data Research | Computer Vision Researcher *Jan 2026 – July 2026*

- Co-authored an autonomous perception study across **22,000+ images**, converting raw data into **Parquet** format and staging on **Hugging Face Hub** to enable high-throughput streaming onto **HPC clusters (Slurm, PyTorch DDP)**
- Applied **Bridged Transfer Learning (BTL++)** with model checkpointing to eliminate simulation artifacts and proved synthetic pre-training reaches saturation at **20–40%** real data, cutting labeling costs by **up to 80%**

Wildfire Predictor (Industry-Sponsored) | Geospatial Machine Learning Engineer *Aug 2025 – Dec 2025*

- Designed a geospatial ingestion pipeline automating preprocessing for **160,000+ spatial and time-series records (HDF, GeoTIFF)** with CRS alignment to feed an interactive UI to validate model outputs as a corporate platform POC
- Engineered a specialized sub-model within a predictive AI fusion architecture, achieving **80% model recall** on early-warning risk classifications to optimize emergency responder resource allocation

Skills

Languages & Frameworks: Python (PyTorch, TensorFlow, Keras, Pandas, NumPy), SQL, C++, Java, HTML/CSS/JS

Tools & Platforms: AWS, Azure, GCP, Git/GitHub, GitHub Actions (CI/CD), Docker, Linux, Slurm, REST APIs, Databricks

Certifications: Amazon Web Services (AWS) Certified Machine Learning Engineer - Associate (Issued Aug 2026)